

WA #5

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Problem #1

a. smallest = 50 because $\text{const int } n = 50$
 largest = 100 because 2 process $\times 50 = 100$

P ₁	P ₂
load	load
add	add
store	store
load	load
⋮	⋮
50 times	50 times

b) smallest = 50
 largest = $50 \times K$

Problem #2 - not atomic

Initial: $s = 1$

Process P₁ starts wait(s)
 reads $s = 1$
 determines $s > 0$ (TRUE)
 gets interrupted

Process P₂ starts wait(s)
 reads $s = 1$
 determines $s > 0$
 gets interrupted

Process P₂ starts wait(s)
 reads $s = 1$
 determines $s > 0$
 decrements $s : s = 0$
 enters critical section

Process P₁ resumes
 decrements $s : s = -1$
 enters critical section

Results P₁ and P₂ are in the critical section at the same time.

Problem #3

```
void spokesperson() {  
    while (true) {  
        wait(question);  
        answer_question();  
        signal(answer);  
        signal(ready);  
    }  
}
```

```
void reporter() {  
    while (true) {  
        wait(ready);  
        ask_question();  
        signal(question);  
        wait(answer);  
    }  
}
```

Problem #4 Deadlock

4 identical resources

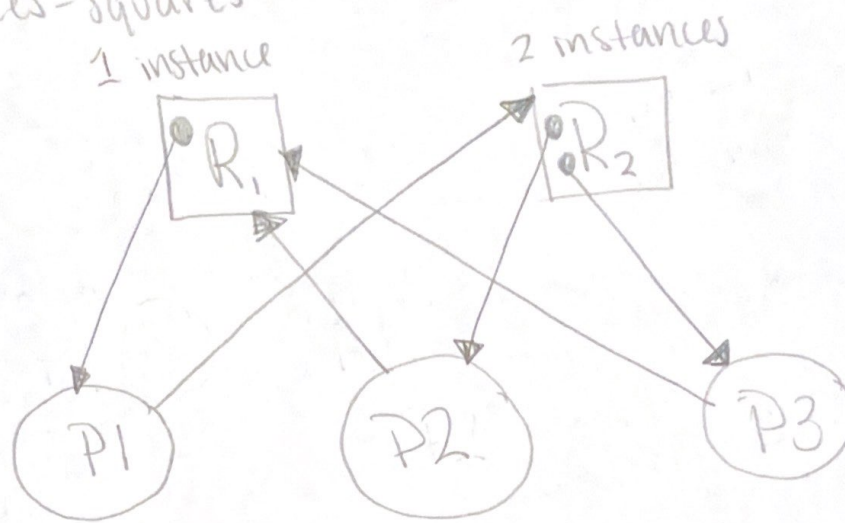
max 2×3 process = 6

deadlock cannot occur because the maximum of 6 exceeds the total available resources which is 4

Problem #5

Kelly Lester

processes - circles
resources - squares



Deadlock occurs because
of a cycle

$P1 \rightarrow R1 \rightarrow P2 \rightarrow R2 \rightarrow P1$